Laporan Pengujian Basis Data Smart Farming

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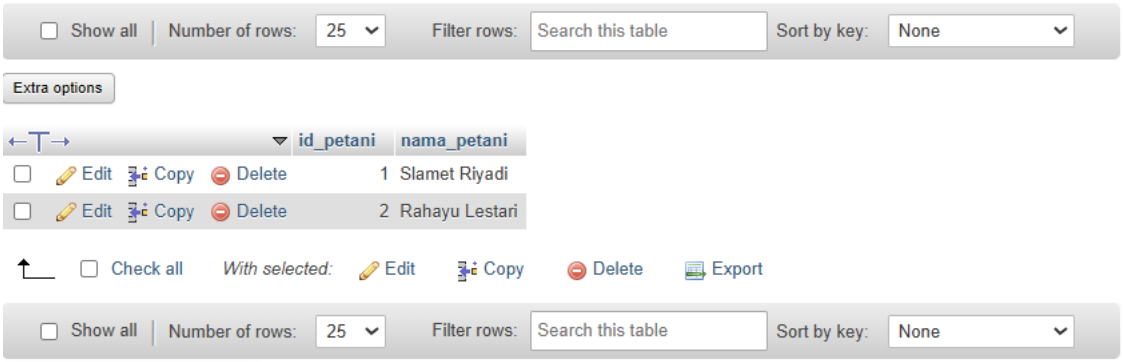
Nim: 264311008

Kelas: 1A.

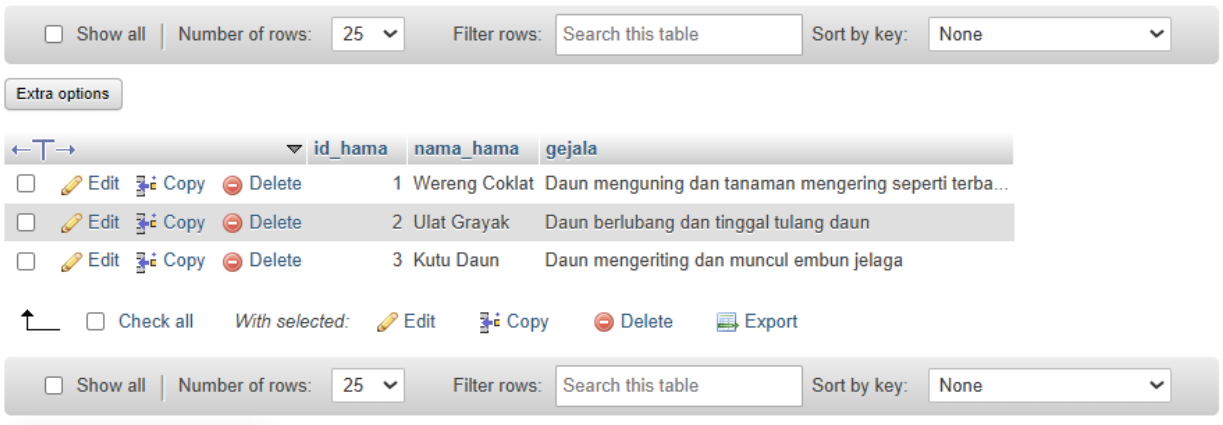
1. Membuat Tabel

Berikut struktur kolom dan isi tiap tabel setelah data uji dimasukkan.

Petani



Hama



Lahan

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_lahan	id_petani	nama_lahan	lokasi	luas_m2	komoditas
Edit Copy Delete	1	1	Sawah Utara	Madiun, Jawa Timur	5000.00	Padi
Edit Copy Delete	2	1	Kebun Cabai	Madiun, Jawa Timur	2500.00	Cabai
Edit Copy Delete	3	2	Ladang Jagung	Ngawi, Jawa Timur	8000.00	Jagung

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

sensor_kelembapan

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_sensor_kelembapan	id_lahan	satuan	status_sensor
Edit Copy Delete	1	1	%	AKTIF
Edit Copy Delete	2	2	%	AKTIF
Edit Copy Delete	3	3	%	NONAKTIF

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

sensor_cuaca

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_sensor_cuaca	id_lahan	status_sensor
Edit Copy Delete	1	1	AKTIF
Edit Copy Delete	2	3	AKTIF

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

sensor_hama

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_sensor_hama	id_lahan	status_sensor
Edit Copy Delete	1	2	AKTIF
Edit Copy Delete	2	3	RUSAK

Check all With selected: Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

pembacaan_kelembapan

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_pembacaan_kelembapan	id_sensor_kelembapan	nilai_kelembapan	waktu_baca
Edit Copy Delete	1	1	65.50	2026-10-26 06:00:00
Edit Copy Delete	2	2	22.00	2026-10-26 06:05:00
Edit Copy Delete	3	1	85.30	2026-10-27 06:00:00
Edit Copy Delete	4	2	55.00	2026-10-27 06:05:00

Check all With selected: Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

data_cuaca

Extra options

	id_data_cuaca	id_sensor_cuaca	suhu	kelembapan_udara	curah_hujan	waktu_baca
Edit Copy Delete	1	1	30.50	70.00	10.00	2026-10-26 12:00:00
Edit Copy Delete	2	2	37.20	60.00	0.00	2026-10-26 13:00:00
Edit Copy Delete	3	1	28.00	90.00	150.00	2026-10-27 15:00:00

Check all With selected: Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

pembacaan_hama

Extra options

	id_pembacaan_hama	id_sensor_hama	tingkat_deteksi	waktu_baca
Edit Copy Delete	1	1	TINGGI	2026-10-27 07:00:00
Edit Copy Delete	2	1	RENDAH	2026-10-28 07:00:00

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

notifikasi_kelembapan

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_notifikasi_kelembapan	id_pembacaan_kelembapan	pesan	waktu_kirim	status_baca
Edit Copy Delete	1	2	Tanah terlalu kering (22.00%). Segera lakukan peny...	2026-10-28 19:17:48	0
Edit Copy Delete	2	3	Tanah terlalu basah (85.30%). Periksa drainase lah...	2026-10-28 19:17:48	0

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

notifikasi_cuaca

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	id_notifikasi_cuaca	id_data_cuaca	pesan	waktu_kirim	status_baca
Edit Copy Delete	1	2	Suhu ekstrem (37.20 C). Waspadai tanaman layu.	2026-10-28 19:17:48	0
Edit Copy Delete	2	3	Curah hujan tinggi (150.00 mm). Waspadai genangan.	2026-10-28 19:17:48	0

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

notifikasi_hama

Show all | Number of rows: 25 Filter rows: Search this table

Extra options

	id_notifikasi_hama	id_pembacaan_hama	id_hama	pesan	waktu_kirim	status_baca
Edit Copy Delete	1	1	2	Deteksi hama tingkat TINGGI di Kebun Cabai. Lakuka...	2026-10-28 19:17:48	0

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table

2. Pengujian yang Ditolak

Ditolak: lahan 99 tidak ada di tabel lahan (melanggar fk_sensor_kelembapan_lahan)

```
INSERT INTO sensor_kelembapan (id_lahan, satuan, status_sensor)
VALUES (99, '%', 'AKTIF');
```

```
SQL Error [1452] [23000]: Cannot add or update a child row: a foreign key constraint
fails (`smart_farming`.`sensor_kelembapan`, CONSTRAINT `fk_sensor_kelembapan_lahan`
FOREIGN KEY (`id_lahan`) REFERENCES `lahan` (`id_lahan`) ON DELETE CASCADE ON UPDATE
CASCADE)
```

Ditolak: kelembapan 150% (melanggar chk_nilai_kelembapan)

```
INSERT INTO pembacaan_kelembapan (id_sensor_kelembapan, nilai_kelembapan)
VALUES (1, 150.00);
```

```
SQL Error [3819] [HY000]: Check constraint 'chk_nilai_kelembapan' is violated.
```

Ditolak: luas lahan negatif (melanggar chk_luas)

```
INSERT INTO lahan (id_petani, nama_lahan, lokasi, luas_m2, komoditas)
VALUES (1, 'Lahan Uji', 'Madiun', -100.00, 'Padi');
```

```
SQL Error [3819] [HY000]: Check constraint 'chk_luas' is violated.
```

Ditolak: status sensor di luar ENUM

```
INSERT INTO sensor_cuaca (id_lahan, status_sensor)
VALUES (1, 'ERROR');
```

```
SQL Error [1265] [01000]: Data truncated for column 'status_sensor' at row 1
```

Ditolak: nama hama duplikat (melanggar uq_nama_hama)

```
INSERT INTO hama (nama_hama, gejala)
VALUES ('Ulat Grayak', 'Gejala lain');
```

```
SQL Error [1062] [23000]: Duplicate entry 'Ulat Grayak' for key 'hama.uq_nama_hama'
```

Ditolak: hama 2 masih dipakai notifikasi_hama (melanggar ON DELETE RESTRICT)

```
DELETE FROM hama WHERE id_hama = 2;
```

```
SQL Error [1451] [23000]: Cannot delete or update a parent row: a foreign key
constraint fails (`smart_farming`.`notifikasi_hama`, CONSTRAINT
`fk_notif_hama_master` FOREIGN KEY (`id_hama`) REFERENCES `hama` (`id_hama`) ON
UPDATE CASCADE)
```